

(P.T.O.)

BTS-IV(R/S)-06-24-2776

		Marks	BL	CO	PO
IV. (a)	Verify Cauchy's theorem for the integral of z^3 taken over the boundary of the rectangle with vertices $-1, 1, 1+i, -1+i$.	6	L2	2	2
(b)	Expand $\frac{z-1}{z^2}$ in powers of $z-1$ and determine the region of convergence.	6	L3	2	3
OR					
V. (a)	Using Cauchy's Residue Theorem, evaluate $\int_c \frac{z}{(z-1)(z-2)^2} dz$ where c is $ z-2 = \frac{1}{2}$.	6	L3	2	3
(b)	Evaluate $\int_0^{2\pi} \frac{d\theta}{2 + \cos \theta}$.	6	L2	2	2
VI. (a)	Find the partial differential equation of all spheres whose centres lie on z -axis.	6	L3	3	2
(b)	Solve $r - 4s + 4t = \sin(3x + 4y)$.	6	L3	3	3
OR					
VII. (a)	Find the partial differential equation by eliminating the arbitrary function from $f(x+y+z, x^2+y^2+z^2) = 0$.	6	L2	3	2
(b)	Solve $x(y-z)p + y(z-x)q = z(x-y)$.	6	L2	3	3
VIII. (a)	Solve $U_{xx} - U_y = 0$ by the method of separation of variables.	6	L1	4	1
(b)	A string is stretched and fastened to two points 1 apart. Motion is started by displacing the string in the form $y = a \sin\left(\frac{\pi x}{l}\right)$ from which it is released at time $t = 0$. Show that the displacement of any point at a distance x from one end at time t is given by $y(x, t) = a \sin\left(\frac{\pi x}{l}\right) \cos\left(\frac{\pi ct}{l}\right)$.	6	L4	4	4
OR					
IX. (a)	Solve the heat equation.	6	L1	4	1
(b)	Solve Laplace's equation in two dimensions.	6	L3	4	3

Blooms's Taxonomy Level's
 L1 - 15%, L2 - 45%, L3 - 25%, L5 - 15%.
